

Observation is Theory-Laden Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 25-35 min

Category	Natural Sciences	Time	25-35 min
Difficulty	Medium	ID	observation-is-theory-laden

Big idea: Seeing scientifically requires concepts.

Student-Facing Prompt

Inspect Observation is Theory-Laden Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Observation is Theory-Laden Stimulus A	Student-facing stimulus 1: The same candle flame described by a poet, chemist, safety inspector, and historian. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Observation is Theory-Laden Stimulus B	Student-facing stimulus 2: A simple phenomenon viewed with different guiding questions. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Observation is Theory-Laden Reveal / Twist	Student-facing stimulus 3: A notice-and-ignore table showing how theory directs attention. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Seeing scientifically requires concepts.

Actual Lesson Questions

#	Question for students
1	After inspecting Observation is Theory-Laden Stimulus A, what is your first knowledge claim?
2	Which exact detail from Observation is Theory-Laden Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Can observation be theory-free?

Ready-to-Use Examples

- The same candle flame described by a poet, chemist, safety inspector, and historian.
- A simple phenomenon viewed with different guiding questions.
- A notice-and-ignore table showing how theory directs attention.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Show an image without context and ask students what they observe.
2. Now label it as a microscope image, satellite image, medical scan, particle trace or climate graph.
3. Ask what new things they notice.
4. Discuss whether the image changed, or the knower changed.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Can observation be theory-free?
- Does training help scientists see more, or see differently?
- What role do models play in scientific knowledge?

Knowledge Questions

- Can observation be theory-free?
- Does training help scientists see more, or see differently?
- What role do models play in scientific knowledge?

Transfer Example

For example, run this activity with the same candle flame described by a poet, chemist, safety inspector, and historian. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...